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Experimental validation of the nerve conduction velocity selective recording technique using a multi-contact cuff electrode

K. Yoshida [a,b,*], G.A.M. Kurstjens [a], K. Hennings [a,c]

[a] Center for Sensory Motor-Interaction (SMI), Department of Health Science and Technology, Aalborg University, Frederik Bajers Vej 7D, DK-9220 Aalborg Ø, Denmark
[b] Biomedical Engineering Department, Indiana University-Purdue University Indianapolis, 723 W. Michigan St, SL-164H, Indianapolis, IN 46202, USA
[c] Mech-Sense, Department of Gastroenterology, Aalborg Hospital, Mølleparkvej 4, DK-9000 Aalborg, Denmark

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ABSTRACT

The earthworm (*Lumbricus terrestris*) is presented as an *in vitro* model of a peripheral nerve containing only two fibers each with distinctly different conduction velocities, the median and lateral giant fibers (MGF and LGF). The worm model is used with a multi-contact cuff electrode to validate the spatial–temporal filtering effect of different electrode contact configurations and the effect of applying a delay adder and matched filter tuned to either the MGF or LGF action potential (AP) to extract conduction direction and velocity from the recording. The results confirmed the known effect of inter-electrode spacing and bipolar and tripolar recording configuration on the AP amplitude. It also demonstrates a crossover point where the amplitude of the tripolar recording is larger than the monopolar recording, an effect of the slower action potential conduction velocities in the worm. The delay adder was found to be an effective velocity sensitive filter, able to discriminate units based on conduction velocity. The matched filter to be an effective means to eliminate artifact and increase signal to noise ratios, however was not found to improve selectivity.

1. Introduction

Different computational models incorporating voltage activated ion channels [1] and the cable equation [2,3] have been developed over the past 50 years [4–7] to understand the biophysics of the electric field captured by an extracellular recording device. Although most models are significantly simplified, they made it possible to simulate a myriad of potential recording configurations and paradigms to give insights into how electrode geometries and processing techniques could improve different types of selectivity of recording. The predictions from these models can next be tested *in vivo* in the acute mammalian model to determine whether they translate to the real world. However, mammalian models can still be complex and their results difficult to interpret. For example, models indicated that fascicle selective recording of nerve activity could be obtained by using a cuff electrode which places multiple electrode contacts at different points along the circumference of the nerve. These predictions were confirmed in [8] to some degree, although the complexity of the model made it difficult to gain insights on what to add to the computational model to further improve the method.

Another type of selectivity, the sensitivity to discriminate neural signals of different nerve fibers of a particular caliber (fiber diameter), or size selectivity, can be optimized by taking advantage of the differences in action potential wavelengths and conduction velocities from different calibers using an appropriate spacing between recording contacts in the direction of nerve conduction [9–11]. By tuning the contact spacing to allow constructive summation of the action potential as a function of wavelength, a spatial–temporal filter can be formed. Multiple, simultaneous observations from multiple locations along the nerve, delayed and summed as suggested by [12] and [13] can be used to create a conduction velocity selective recording paradigm. Such filters are optimized on the conduction velocity making them velocity selective filters (VSF). This class of filter is a Delay and Sum-Beamformer, and has been used in sonar [14] and ultrasound [15] applications. In [13], it was first proposed to use a multi-contact cuff electrode to form multiple tripolar configurations [9] for velocity selective recording (VSR). It described in theory the method of using an array of double differential amplifiers, artificial time delays for each selected velocity, an adder and narrow band filters. Theoretically, selectivity could further be enhanced by passing the raw signal through an optimal match filter [16] before entering the beamforming filter.

* Corresponding author at: Biomedical Engineering Department Indiana Univ - Purdue Univ Indianapolis 723 W. Michigan St., SL220F Indianapolis, IN 46202, USA. Tel.: +1 (317) 274 9714; fax: +1 (317) 278 2455.
E-mail address: yoshidak@iupui.edu (K. Yoshida).

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A major difficulty in the process has been the experimental validation of the theoretical prediction. The first *in vitro* experiments in frog demonstrated evidence of feasibility but managed to identify the conduction velocity of only the dominant fiber group [17]. Recognizing that a larger rejection of stimulus artifacts and a reduction in electrode and amplifier noise were needed, a dedicated low-noise amplifier system was designed [18], which made it then possible to increase the number of fiber populations that could be distinguished [19]. Nevertheless, for use with naturally occurring nerve activity the selectivity needs to be increased further. Enlargement of the delay adder, however, increases the detection threshold as a larger amount of nerve signal power is needed to exceed the increased noise power [20].

Previous experimental validations were limited to electrically evoked compound action potential (CAP) response. Ideally, the optimal and beam forming filters should be trained with data recorded from identified single nerve fibers. However, the process of collecting a set from an amphibian or mammalian nerve bundle that can contain hundreds–thousands of nerve fibers and ensuring that each identified unit is a single unit is a daunting technical challenge. Moreover, because of the relatively high conduction velocities of mammalian nerves, the length of nerves used in previous *in vitro* experiments are short, leading to (1) an overlap in stimulation artifact and CAP response and (2) synchronous contributions from different fiber groups which cannot be individually recognized. Because of these two factors, it is difficult to compare the effect of VSFs of different fiber groups from the CAP.

In the present study, we used a new experimental model, *Lumbricus terrestris*, the earthworm as a representative model of an *in vitro* peripheral nerve preparation. A multi-contact cuff electrode was placed around the entire worm, which modeled a peripheral nerve. This "nerve" contains three myelinated giant fibers, one median giant fiber (MGF) and two lateral giant fibers (LGF) that are located in a ventral nerve cord (VNC) which extends through the entire length of the earthworm [21,22] and have a nerve conduction velocity ranging 15–45 m/s and 7.5–15 m/s respectively [23–25]. These giant fibers mediate rapid withdrawal reflexes of the anterior (MGF) and posterior (LGFs) [26,27]. When action potentials are elicited in the LGFs they are conducted synchronously as the two LGFs electrically coupled by collateral branches and functions as a single unit. The model is representative to the traditional whole nerve cuff electrode recording technique, except that the worm model can be simplified to a nerve containing only two fibers each with distinctly different conduction velocities, which were used to experimentally test two VSF techniques.

2. Methods

2.1. Experimental set-up

The earthworms (*L. terrestris*) used in the present study were purchased from a local supplier (Hvidovre Sport, Hvidovre, Denmark) and kept quiescent at 4 °C. 24 h before the experiments, when they were moved to a cabinet and kept at room temperature (21 °C).

The experimental set consisted of work on twelve adult earthworms. The earthworms were anesthetized in a 10% ethanol solution for 10 min aerated with ambient air. Following the anesthesia the posterior end of the earthworm was threaded through the multi-contact cuff electrode with the aid of a suture. The multi-contact cuff electrode, unlike those used in [28,29], had an inner diameter of 4 mm and contained 11 equally spaced platinum foil ring contacts (0.5 mm wide, 3 mm contact spacing). Subsequently, the earthworms were placed in a Sylgard® bottomed Petri dish which had two additional blocks of Sylgard® (1 cm in height) on which the head and tail of the worm was placed (see Fig. 1). The head and tail of the worm were pinned to the blocks of Sylgard® with two sets of needle electrodes. These two sets of needle electrodes were used for stimulating the anterior and posterior end of the worm, respectively. The multi-contact cuff electrode was placed on the bottom of the Petri dish, which was filled with normal isotonic saline (0.9% NaCl).

Signals from the electrode were differentially amplified and filtered (gain: 1000×, highpass: 0.1 Hz). The two outermost electrode contacts were shorted and served as the reference (see Fig. 1). Adequate stimuli of the MGF and LGF was accomplished using standard 100 μs long rectangular, current controlled pulses (SD9 + PSIU6, Grass-Telefactor, USA). The resulting ENG signals were sampled at 50 kHz. The inter-stimulus interval was 2 s. The experimental acquisition and control was accomplished using a PC with a PCMIO-E1 card (National Instruments, USA) running experimental control and data acquisition software (MrKick, Aalborg University, Denmark).

2.2. Protocol

First, the thresholds intensities (minimum intensity evoking a response from each of the fibers) were established and the supramaximal stimulation intensity, the level that would activate steady both the MGF and LGF fibers, was set to two times the threshold found. Then, the action potentials elicited by at least 50 stimuli to the head of the earthworm were recorded. In six experiments, the tail of the earthworm was subsequently stimulated and the responses to at least 50 stimuli were recorded.

2.3. Signals processing

2.3.1. Electrode spacing

The experimental data were recorded with each channel of the 8 channel amplifier configured to amplify adjacent rings (3 mm electrode spacing) of the multi-contact cuff electrode in a differential bipolar recording configuration. In the data analysis these data were digitally post-processed to obtain all other possible bipolar and tripolar configurations with electrode spacings of 6 mm, 12 mm, 18 mm, and 24 mm. These signals were obtained from the original bipolar signals, where the bipolar channel number n with an electrode spacing of 3 mm ($b_{3\text{mm},n}$) can be written as $m_n - m_{n-1}$ (m denoting a monopolar channel, and n the channel number). Consequently, the bipolar channel $b_{6\text{mm},n}$ can be found as $b_n + b_{n-1} = m_n - m_{n-1} + m_{n-1} - m_{n-2} = m_n - m_{n-2}$. The bipolar signals with 12 mm, 18 mm, 24 mm were obtained by adding 1, 2, and 3 additional 3 mm bipolar channel to this sum, respectively. Once all of the bipolar signals were found, the tripolar signals were calculated from the derived bipolar signals.

2.3.2. True conduction velocities

The "true" conduction velocities of the MGF and LGF units where found from the seven tripolar signals (3 mm inter-electrode spacing). The middle negative peak in each recording indicates when the action potential passed the center electrode of the 3 mm tripolar triple. This instance was identified for each of the seven channels to give the time and position of the unit as the action potential traversed the cuff electrode. A linear least squares regression was made on the time vs position data for each of the two units, and their "true" conduction velocities were estimated as the slope of the regression.

2.3.3. Filters

The VSF tested in the study was implemented by delaying each 3 mm tripolar channel by a multiple of a constant delay (τ) and the channel number (n), and following this with a summing operation of the delayed outputs (see Fig. 2). The delays were transformed to

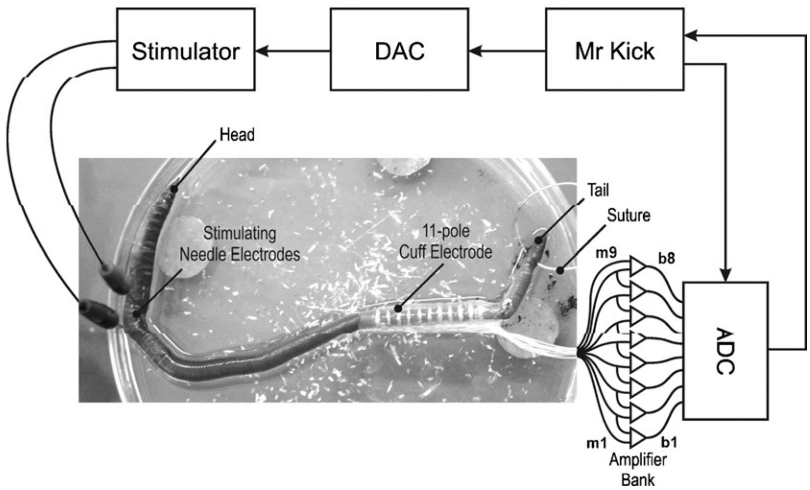

Fig. 1. Experimental layout illustration showing a photograph of the worm in a Sylgard® lined experimental dish, stimulating electrodes, 11 pole multi-contact cuff electrode and a block diagram of the surrounding experimental instrumentation. The stimulating electrodes were placed, in this experiment, at the head of the worm after the worm was threaded into the multi-contact cuff electrode using a suture placed through the tail of the worm. The experiment was controlled by Mr Kick, an experimental control and data acquisition software, which triggered an opto-isolated stimulator (Grass SD9 + PSIU6). Bipolor records b_1–b_8 from the 9 inner poles m_1–m_9 of the 11 pole electrode were made using a custom built multi-channel amplifier. The two outermost poles were shorted and connected to the signal ground (not shown).

conduction velocities (CV) based on the electrode distance and the delay.

To further enhance the signal to noise ratio (SNR), the output of the delay adder was passed through a matched filter tuned to either the MGF or LGF unit. To create and test the matched filters, the data set was divided into training and test data sets. The training set was used to create the spike templates for the each fiber and consisted of the first 25 sweeps of the original data set. The test set consisted of the remaining sweeps. Thus, the training set contained at most 50% of the sweeps recorded in an experiment. To create the spike templates, a region-of-interest (ROI) was identified. The spikes were extracted, aligned by their peaks, averaged and time-reversed to create the impulse response of the matched filter. The filter was implemented by convolving the impulse response of the filter with the output signal from the delay adder (see Fig. 2).

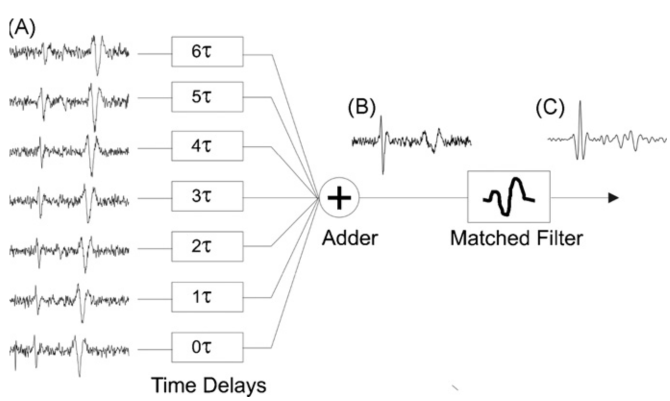

Fig. 2. A schematic representation of the velocity sensitive recording stages used in the experiments. (A) shows seven 3 mm tripolar records synthesized from eight simultaneously acquired bipolar records from nine rings on a 11 polar cuff electrode placed around the earthworm. The traces are resulting from a single suprathreshold stimulus delivered through needle electrodes placed at the head of the worm, and thus the noise seen on each channel is representative of an unfiltered single sweep record. The resulting caudal conduction of the action potentials in the two giant axons is captured by the multiple rings of the cuff electrode. The traces were processed by passing them through a bank of delay filters and summed to form a single delay adder filtered trace (B). The delays were tuned, in this case, to the conduction delay of the first spike (from the MGF). Finally, a matched filter, also matched to the MGF, was applied to result in the matched filtered trace (C). Note in the raw signal, the MGF (1st spike) amplitude is smaller than the LGF (2nd spike). Following delay adder filtering, the MGF spike is amplified, while the LGF spike is attenuated. The matched filter further increases this differentiation.

2.4. Analysis

2.4.1. Effect of filtering

The data set originates from experiments conducted on 12 earthworms. Eight simultaneously recorded traces from the eight individual bipolar channels of the multi-contact cuff electrode comprise a data sweep. 44 sweeps from each of the 12 experiments (n = 528) comprises the data set.

Four types of filters can be defined given the two types of filters, delay adder and matched filter, depending upon which unit they are tuned to, the MGF or LGF. The delay adder (DA) filtered responses that were tuned to the MGF are denoted as DA_mgf, while the DA matched to the LGF, DA_lgf. Similarly, the two respective combinations for the matched filter (MF) filtered responses were the MF tuned to the MGF (MF_mgf), and the MF tuned to the LGF (MF_lgf). Each sweep was collected pre-triggered to the stimulus pulse, so that 500 points were collected before and 1500 points after the stimulus (20 ms total duration). Four regions-of-interest in time (ROI) were defined for each sweep. Three of these periods defined the location in time where deflection from the stimulus artifact (SA) and the MGF and LGF action potentials were present. The fourth period was the first 500 points, sampling the background noise in the record.

The peak to peak amplitude (Vpp) was measured in each of the four defined periods to quantify the amplitude of the SA, MGF action potential, LGF action potential and background noise for the raw signal (Raw) and the four filter types (DA_mgf, DA_lgf, MF_mgf, MF_lgf).

The SNR for each type of filtering was estimated by taking the response Vpp and dividing it by the noise Vpp and subtracting 1 from each sweep under the 10 filter/ROI combinations. Finally, the relative amplitudes of the tuned to not-tuned responses were quantified by dividing the tuned response SNR by the not-tuned response SNR.

Statistical difference between the Vpp response at the different ROIs and using the different filters was assessed using a within design two way repeated measures analysis of variance (ANOVA) and the Neuman–Keuls multiple comparisons post hoc analysis. The same repeated measures ANOVA and post hoc test was used

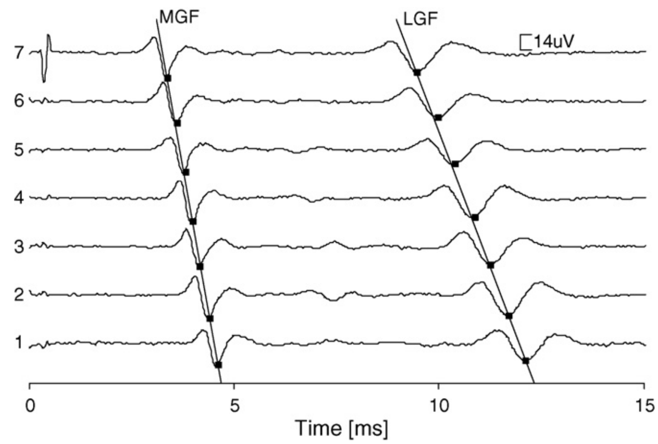

Fig. 3. Action potentials recorded with a tripolar configuration by the multi-contact cuff electrode (averaged fifty times). The cuff electrode were placed around the entire earthworm and the intensity of the stimuli was set to a level that would evoke an action potential in both the median giant fiber (MGF) and the lateral giant fiber (LGF). The conduction velocities of the MGF and LGF were estimated from the conduction delays between the peaks of their action potentials (■ peak of the action potential). In this experiment the conduction velocities were 14.9 m/s and 6.9 m/s for the MGF and LGF, respectively.

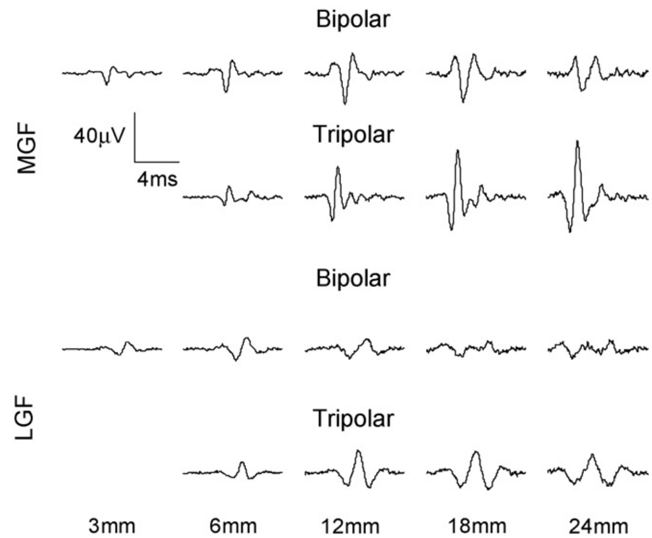

Fig. 4. Example of signals vs effective cuff length. The effect of increasing the contact spacing of the cuff electrode, i.e. the effective cuff length, on the amplitude and waveform the extracellular potential of a median giant fiber (MGF) and a lateral giant fiber (LGF) from the same experiment. The fibers had a conduction velocity of 16.9 m/s and 6.3 m/s for the MGF and LGF, respectively. These extra-cellular potentials were reconstructed from the eight bipolar recordings that all had a contact spacing of 3 mm. The procedure for reconstructing these recording configurations is described in Section 2.

separately for assessing the SNR and tuned to not-tuned response. The statistical significance level was set to $\alpha = 0.05$ in all cases.

3. Results

3.1. "True" conduction velocities

A typical stimulus triggered average record from an experiment is shown in Fig. 3. In the tripolar recording configuration, the action potential appears as a triphasic waveform as it passes by the three poles of the recording electrode set. Each recording tripole shows a slightly delayed deflection, which corresponds to the conduction delay of the action potential. In Fig. 3, the MGF action potential first arrives at electrode set 7 and sequentially shows deflections in set 6, 5, and so on until set 1. The action potential from the LGF arrives at electrode set 7–6 ms after the MGF and sequentially shows deflections in the other electrode sets, but at increasing conduction delays due to its slower CV.

The mean CVs from all the experiments was found to be 15.6 ± 0.87 m/s and 7.7 ± 0.48 m/s for the MGF and LGF respectively (mean ± standard error of mean (SEM), $n = 12$).

3.2. Effect of inter-electrode spacing on the signal

The amplitude of the action potential is influenced by, among other factors, the inter-electrode spacing and the CV of the action potential. Fig. 4 shows an example of the same MGF and LGF action potential recorded with different electrode contact spacing in the bipolar and tripolar recording configuration for a single experiment.

Two generalizations can be made. Increasing the inter-electrode spacing increases the amplitude of the action potential as recorded from the contacts until the spacing approaches the wavelength of the action potential and the action potential no longer adds constructively. This occurs at about 18 mm for the LGF. Second, the tripolar configuration has larger amplitude than the bipolar configuration. The average relationship of the peak to peak action potential amplitude as a function of inter-electrode spacing across all animals is shown in Fig. 5 and summarizes the two observations made above.

It shows, moreover, that the maximum tripolar amplitude is different for the two fibers. The tripolar LGF has maximum amplitude an inter-electrode spacing of 12 mm, while the faster MGF is still increasing in amplitude at 24 mm. The figure also shows at smaller inter-electrode distances, the bipolar configuration has larger amplitude than the tripolar configuration. The crossover points where the tripolar configuration amplitude begins to exceed the bipolar configuration amplitude are ∼6 mm for the LGF and the ∼8 mm for the MGF.

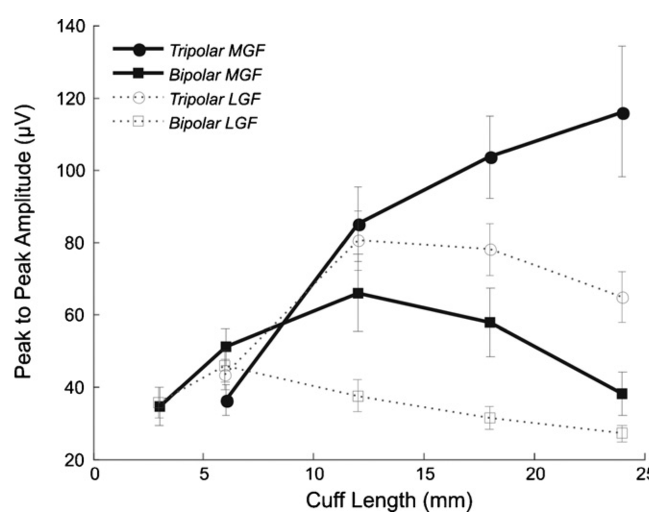

Fig. 5. Peak-to-peak amplitudes of the extracellular action potential of the median giant fiber (MGF, solid lines) and lateral giant fiber (LGF, dotted lines) as a function of effective cuff length. These recordings were reconstructed from eight 3 mm bipolar recording channels to synthesize different contact spacing (6 mm, 12 mm, 18 mm, and 24 mm) and different recording configurations (bipolar (square symbols) and tripolar (circular symbols)). Unlike the original Stein plots in the cat, the tripolar configuration has larger spike amplitude than the bipolar configuration. This is due to the slower conduction velocities and shorter wavelength of the action potentials in the worm (see text).

3.3. Effect of filtering

A typical set of experimental traces is shown in Fig. 6, which qualitatively summarizes the type of signals that were recorded, and the effect of the various types of filters on the spikes. The bottom set of traces shows the raw simultaneous recordings from the multi-polar cuff electrode in the 6 mm tripolar recording configuration. The records show the response to stimulation either at the head of the worm, generating caudal conduction of the two units, and the response due to stimulation at the tail of the worm, generating rostral conduction. Note, as in Fig. 3, the shape of the units remains conserved as it propagates through the cuff electrode, and the response of the MGF has a smaller time shift between recording successive recording channels as compared to the LGF, due to its faster CV. The third column of the figure shows the arithmetic sum of the rostral conducting and caudal conducting traces, and was used to see the effect of an interference waveform on the filters.

The middle and top sets of traces are selected responses following filtering with the delay adder (middle) and delay adder + matched filter (top). The unit highlighted by the solid (MGF) and outlined (LGF) rectangles indicate the unit that the filter was tuned to. Note that the unit that is tuned to is amplified, while the untuned unit/direction is either unchanged or attenuated in amplitude. The absolute amplitude of the signals is also influenced by filtering. The MF strongly attenuates the amplitude of both the signal and noise, with the noise generally attenuated more than the unit. Another notable observation is that regarding the stimulus artifact. If there is an artifact on one of the 7 channels of the raw record, then it is propagated into the delay adder filtered response. Though it is present in the signal that is passed to it, the artifact is eliminated by the MF.

3.3.1. Delay adder filter vs the matched filter

The effect of the DA filter is summarized in Fig. 7, which shows the tuning curve response of the filter as a function of the number of adjacent tripoles used in the DA filter. Each tuning curve trace is generated by taking an example raw data sweep set and passing it through the DA filter at different time delay factors. The amplitude of the resulting filtered data is quantified by measuring its peak to peak amplitude and is plotted as a function of the time delay used. The different traces ordered by increasing line weights are the tuning curve response for using increasing numbers of tripoles, starting with 2 tripoles (9 mm total cuff length) and increasing to 7 tripoles (24 mm total cuff length). The delay adder can be tuned to pass units propagating at a specific CV. The dotted lines show the conduction velocities that the delay adder was tuned to with the higher conduction velocity delay adder tuned to the conduction velocity of the MGF, and the slower conduction velocity delay adder tuned to the LGF. The figure also shows little difference in the velocity sensitivity response between the MF and the DA filter indicating the velocity sensitivity is strictly due to the DA filter. A second observation is with respect to the effect of different cuff lengths. Longer cuff lengths increase the velocity selectivity of the filter.

The tuning curve also predicts that the DA filter should have directional selectivity. Action potentials with a negative CV, travelling in the opposite direction than the tuned direction of the filter, will not be enhanced.

The Vpp of the signal (neural unit), the background noise, and the SA from all 12 experiments were analyzed to quantify the qualitative observations made in above. Fig. 8 is a summary of the grand means ± SEM across the 12 experiments and 3 filtering conditions (Raw (unfiltered), Delay Adder Filtered, and Match Filter Filtered). The grand mean raw amplitude of the MGF and the

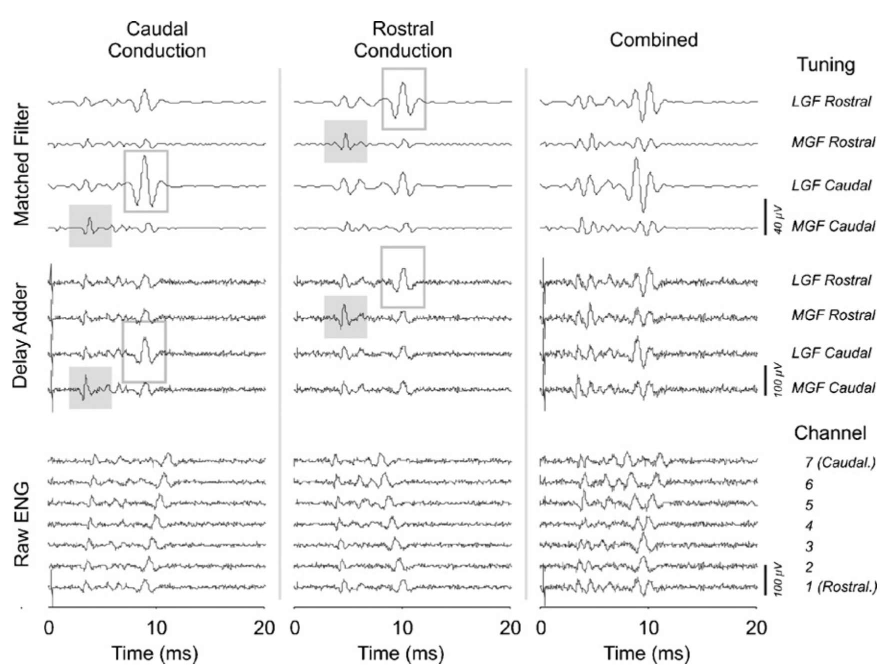

Fig. 6. Example of filtering and conduction direction of the action potential. Three different test conditions are shown. The first column shows the response following stimulation applied to the head of the worm to elicit head–tail conduction (Caudal conduction). In the second column stimulation was applied to the tail of the worm resulting in tail–head conduction (Rostral conduction) in the giant axons. The third column shows the sum of the Rostral and Caudal conduction cases is the interference waveform simulating the superimposed conduction of four fibers. The bottom set of traces is the seven raw simultaneous tripolar records from the multi-contact cuff electrode, where the third column interference waveforms are created by the addition of the raw data sets in the first and second columns. The middle set of traces is the result following delay adder filtering for four tuning cases: Rostral conducting LGF, Rostral conducting MGF, Caudal conducting LGF, and Caudal conducting MGF. The top set of traces is the result following matched filtering for the four tuning cases used in the delay adder. The highlighting squares in the first two columns highlight the action potential to which the filters were tuned. For clarity, the MGF highlight is solid. Generally, the direction/unit tuned spike is amplified by the filtering, while the untuned spike is attenuated in the non-superimposed case. In the combined case, however, the differences are less clear.

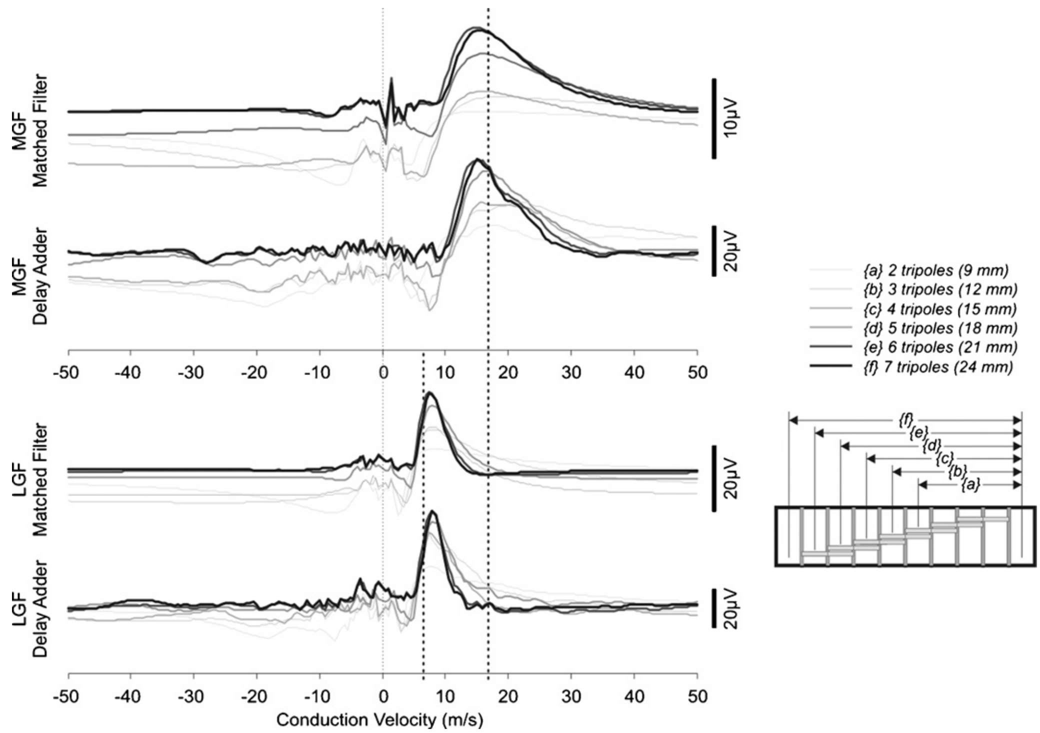

Fig. 7. Filter tuning curves for a typical LGF and MGF unit. The tuning curve was generated from a single randomly selected example sweep with both units active. The peak to peak values of the unit filtered through a filter tuned to a particular conduction velocity are shown. The small inset below the legend is a representation of the multi-contact cuff electrode and the seven 6 mm tripoles it contains. The 7 sets of adjacent tripoles used in the DA { }, the number of adjacent tripoles used, and cuff lengths in ( ) are also indicated. Four panels showing the tuning curves for the two units (LGF and MGF) and two filters (delay adder and delay adder + matched filter) are shown. The series of curves for each panel are tuning curves using increasing numbers of adjacent 6 mm tripoles. The dotted lines indicate the "true" conduction velocity measured for the faster MGF unit and the slower LGF unit. The curves show the highest amplitudes are found when the filter is tuned to the conduction velocity and correct conduction direction of the unit. The matched filter does not appear to add further velocity or direction discrimination. Finally, the faster unit (MGF) appears to require a longer cuff to obtain larger peak values.

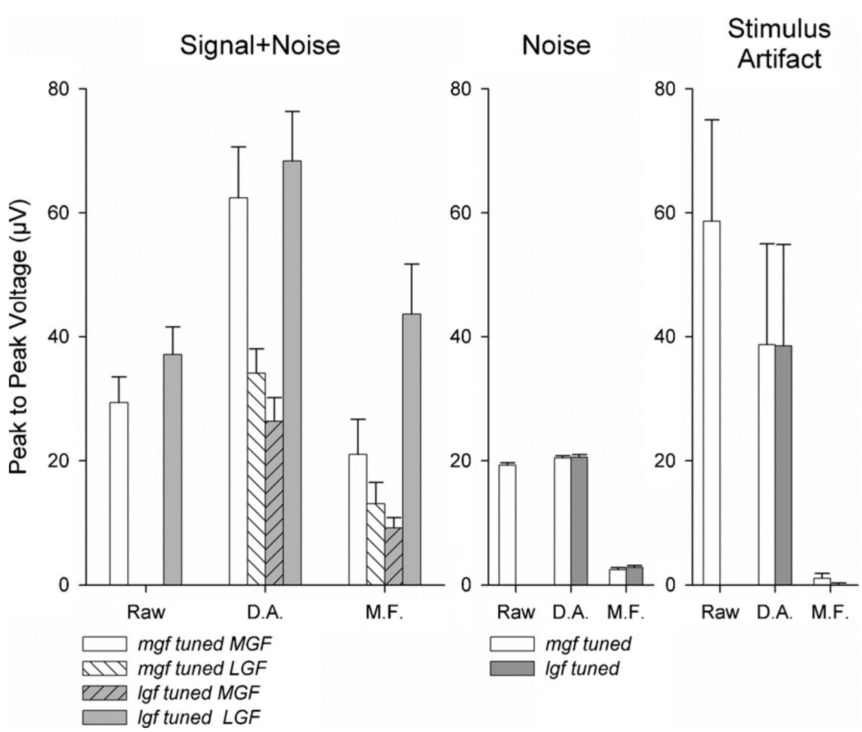

Fig. 8. Peak to peak amplitudes: The grand mean ± standard error of mean (SEM) ($n = 12$) across all experiments of the peak to peak unit amplitudes compared with the amplitude of the baseline noise and stimulus artifact as a function of filtering. The MGF related amplitudes are with empty bars, while the LGF related amplitudes are with gray bars. Hatched bars indicate unit amplitudes measured with filters tuned to the other unit. In the left panel, the use of the delay adder (DA) filter significantly increases the amplitude of the unit the filter was tuned, while not influencing or decreasing the amplitude of the other unit. The matched filter (MF) generally decreases the absolute amplitude of the DA signal, while maintaining the differentiation the MF established. The background noise (middle) and stimulus artifact (right) are not altered by the DA filter, but are significantly reduced by the MF.

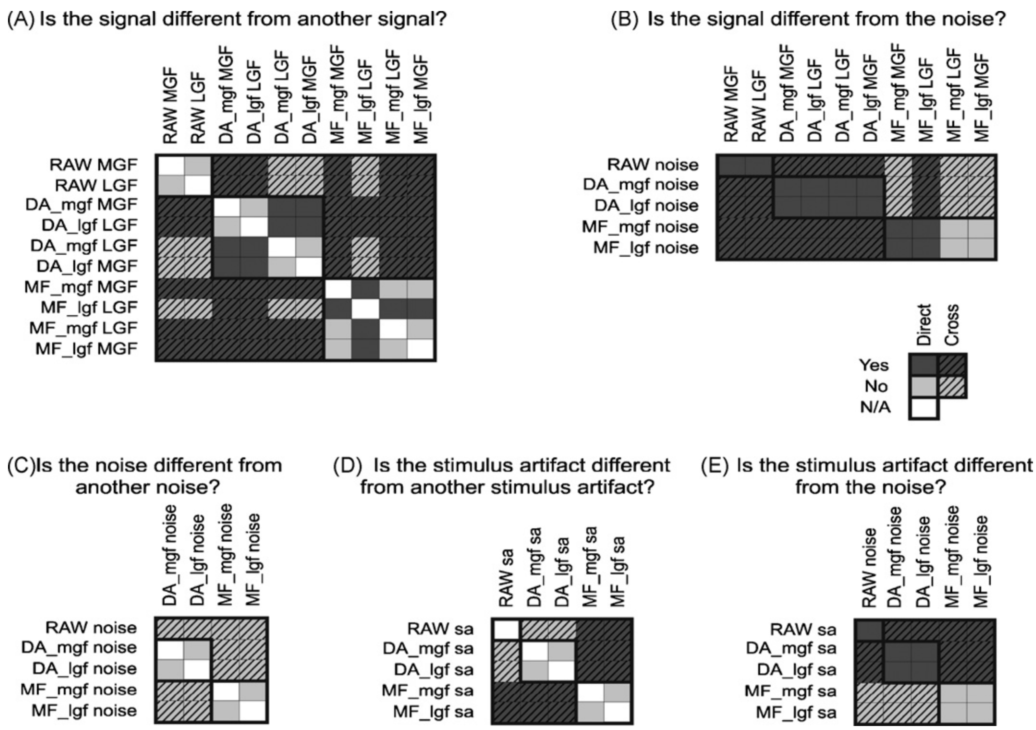

Fig. 9. Amplitude comparisons: The results of a three way repeated measures ANOVA and post hoc tests on the amplitude of the signal, noise and stimulus artifact. The two way interaction outcomes of the comparisons are shown. Significant differences ($\alpha < 0.05$) are marked as dark gray, while non-significant differences are marked as light gray. Comparisons made within a filter type are denoted as Direct comparisons, while comparisons made between filter types are denoted as Cross comparisons and are indicated by hatched squares. The labels on the table indicate the signal conditioning type followed by what was filtered using the following abbreviations:
RAW = no filtering,
DA_mgf = delay adder tuned for the MGF,
DA_lgf = delay adder tuned for the LGF,
MF_mgf = delay adder + matched filter tuned to the MGF,
MF_lgf = delay adder + matched filter tuned to the LGF,
MGF = unit from the medial giant fiber,
LGF = unit from the lateral giant fiber,
sa = stimulus artifact.
(A) shows that the delay adder filter increases the difference between the MGF and LGF. The addition of the matched filter increases the difference of the classified units. The non-tuned units remained different from noise unless the matched filter was applied (B). The relationship between noise itself did not change by filtering (C). The stimulus artifact was not altered by the DA, but was altered by the addition of the MF (D), which brought its amplitude down to the level of noise (E).

LGF were 29.4 ± 4.1 and $37.1 \pm 4.4\,\mu V$ respectively. The background noise of the raw signal was $19.3 \pm 0.4\,\mu V$. The DA filter amplified the matched unit amplitude to 62.4 (MGF) and 68.4 (LGF) μV, while the unmatched unit amplitude was unchanged from the raw signal at 34 (MGF) and 26 (LGF) μV. The SA was reduced on average from 58.7–38.7 μV with the use of the DA filter. The addition of the MF resulted in an average reduction in the amplitude of both the signal and the noise to 21.0 μV (MGF signal) and 43.6 μV (LGF signal), and 2.4 μV (MGF noise) and 2.8 μV (LGF noise). The SA was reduced in amplitude to 1.4 μV (MGF) and 0 μV (LGF). On average, the slower unit, the LGF, had the tendency for having the larger signal amplitude. This is likely due to its peculiar physiology. The LGF is comprised of two gap-junctioned fibers acting as a single unit, while the MGF is a single fiber.

The results of the ANOVA and two way interactions of the type of filter (Raw, DA, MF) vs the ROI (MGF, LGF, SA, background noise) are summarized in Fig. 9. The direct comparisons lie along the diagonal of the matrices, while comparisons between different types of filtering are off the diagonal and are indicated by diagonal hatching. Fig. 9A shows that the amplitude of the MGF and LGF was not significantly different from each other for the Raw and DA (tuned to the unit), DA (not tuned to the unit), and the MF (not tuned to the unit). The amplitude of the MF tuned to the MGF and detecting the MGF was different to that of the MF tuned to the LGF and detecting the LGF. There was also no difference between the amplitude of the DA (not tuned to the unit) and the amplitude of the Raw unit. Fig. 9B shows that the signal, whether matched or not matched to the filter is different from the noise in all cases except in the case of the MF, where the unmatched unit is not significantly different from the noise. Fig. 9C shows that the noise between different filters and their tuning is not different from one another. Fig. 9D shows that filter tuning does not affect the amplitude of the stimulus artifact, though in the case of the MF, the stimulus artifact is different than the stimulus artifact from the Raw signal and the DA. Finally Fig. 9E shows that the Raw and DA stimulus artifact is different from noise, while in the case of the MF, the stimulus artifact is statistically not different from the noise.

The SNR was estimated from the peak to peak magnitudes of the Raw, and DA and MF processed signals. Processing the signal using a MF (Figs. 8 and 9) results in a significant decrease in the signal amplitude, but the noise also changed. Similarly, the amplitude of the response of the tuned unit from the DA increased relative to that of the Raw signal, while the noise amplitude was found not to be significantly altered. Fig. 10 summarizes the change in SNR, and the results of their statistical interactions test. The DA significantly increases the SNR for the LGF, while the level for the MGF showed an increase, but not sufficient for statistical significance. There was a significant increase in SNR with the use of the MF, where all combinations of the MF were different from one another.

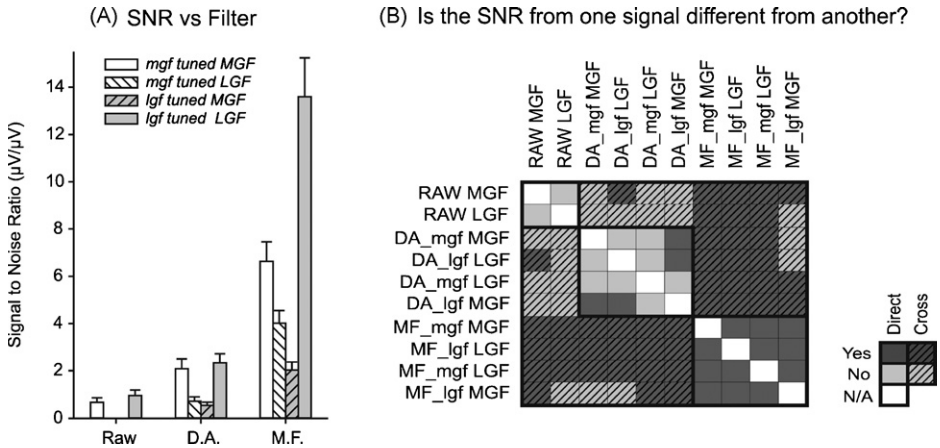

Fig. 10. SNR comparisons: (A) is the grand mean ± SEM. (n = 12) of the signal to noise ratio as a function of filter. It shows that the use of the DA and DA + MF filters has a trend to increase the signal to noise ratio of the tuned unit. Statistical comparison using a three way repeated measures ANOVA indicates that the improvement in the signal to noise ratio is significantly different ($\alpha < 0.05$) only for the delay adder + matched filter configuration.

Fig. 11 shows that the relative difference between tuned and untuned units increases with the use of the DA, but remains, in general, unchanged with the inclusion of the MF. A generalization can be drawn based on these results. The DA filter is the main filtering stage responsible for increasing the velocity based discrimination of the units, while the MF significantly decreases the noise content of the recording. This observation is consistent with theory, as the caliber of the unit does not alter the time course of the action potential recorded by a set of electrodes. As a result, the units are generally temporally similar in shape, though the amplitude and spatial distribution of the action potentials are different for the two units.

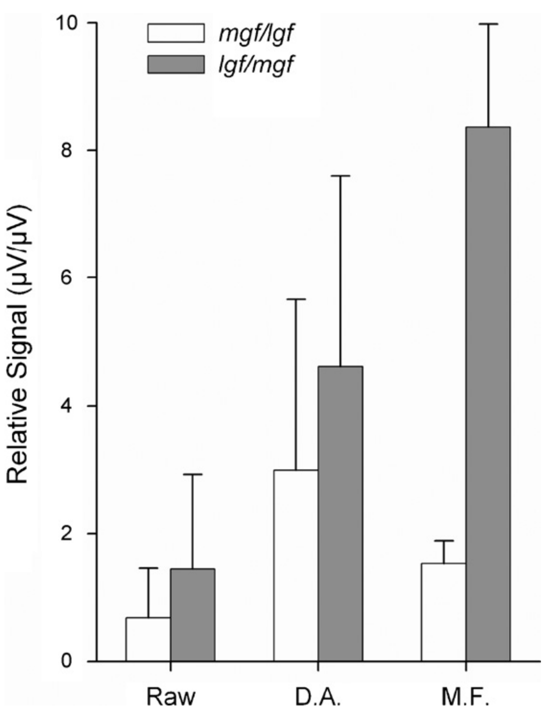

Fig. 11. Relative signal to noise comparisons: The average relative signal to noise ratios from all experiments (n = 12) of the tuned unit/not-tuned unit is shown. In the case of the Raw signal, the two bars are reciprocal of one another. The delay adder increases the difference in the tuned to not-tuned unit by approximately a factor of 2–3. Although the addition of the matched filter increases the signal to noise ratio, its effect on the separation between tuned and not-tuned unit, based on signal amplitude, had a mixed result, improving the relative signal of the LGF, while decreasing that of the MGF.

4. Discussion

The earthworm was presented as an *in vitro* model of the peripheral nerve, and was used to test a multi-channel recording technique based upon beam forming and matched filters using a multi-contact cuff electrode.

4.1. Earthworms: an animal model of in vitro peripheral nerve preparations

We have used the two giant fibers, MGF and LGF, to determine whether the multi-channel recording and signal processing technique can discriminate two units with different conduction velocities, one with approximately half the conduction velocity of the other. The earthworm can be viewed as an inexpensive, readily available alternative that serves an intermediate platform between computer models and *in vivo* testing in mammalian models. Since real electrodes are used, many of the potential issues of noise, recording stability, current loops, and artifacts are present in the set-up, and can be used to debug electrodes and instrumentation being developed for advanced peripheral neural prosthetics, as well as train students and biomedical engineers before moving them on to animal models or human subjects.

L. terrestris where used as a model of a simplified *in vitro* peripheral nerve preparation. Anatomically, the earthworm giant fibers are a series of nerve cells; one for each body segment of the earthworm. These nerve cells are separated by oblique septa [22] and are electrically coupled by gap junctions [30]. Consequently, they function as a single giant fiber that extends through the entire ventral nerve cord. Each of the giant fibers is wrapped in myelin sheaths that resemble the myelin sheaths of vertebrate nerve fibers, with some structural differences. Despite these differences, the myelin sheaths of MGF fibers and vertebrate myelinated nerve fibers share the same functional role and have saltatory conduction [21]. Thus the earthworm models the body of the peripheral nerve having only two nerve fibers, as compared to hundreds to thousands of fibers in a comparably sized mammalian nerve.

4.2. Conduction velocity and inter-electrode spacing

The conduction velocity of the MGF and LGF are considerably slower than that of the mammalian myelinated axon. We measured velocities of 15.6 ± 0.87 m/s and 7.7 ± 0.48 m/s for the MGF and LGF respectively. These values are about half of the values reported by [26] but closer to measured by others [25,27]. The difference may be due to the ethanol used to anesthetize the animals, but does not pose a major problem since the conduction velocities were stable. In part, the slower conduction velocity is due to the temperature at which the experiment was carried out, 21 °C. In part, the difference is due to the physiology of the earthworm axon. Translation of the results to the mammalian nerve requires rescaling the results by the relative wavelength of the action potential. The electrode spacing for mammalian nerves conducting at 60 m/s would be ∼4× longer than the results at 15 m/s. A similar study in the mammalian nerve would require a considerably more difficult protocol involving the identification of a single unit and triggered averages of that unit in the multi-polar recordings, or isolation of a unit through splitting of the filament.

The results showed the effect of inter-electrode spacing on the amplitude and recording configuration. For most of the inter-electrode spacing tested, the tripolar recording configuration showed a larger amplitude than the bipolar configuration for both MGF and LGF action potentials. [9] reported opposite results. This difference could be explained by the difference in conduction velocities of the units tested in the two studies. [9] made their characterization using myelinated fibers from the cat sciatic nerve with conduction velocities ranging from 21 m/s to 64 m/s at 37 °C. The two fibers in our study were considerably slower with mean velocities at 15.6 m/s and 7.7 m/s. Although the inter-electrode spacings were comparable between the two studies, the wavelengths of the action potentials were not. The behavior reported by [9] occurs in the region below the crossover point. Theoretically, a crossover point should exist for the faster mammalian nerve fibers, however, the cuff electrode lengths necessary to reach the crossover point would be ∼4× longer than those used in Stein's original study and not practical for application. Conversely, the inter-electrode spacing necessary to reproduce [9] in our study would be ∼4× shorter than the ones used. Such short cuffs would be impractical to construct. Moreover, the amplitude of the signal would diminish below the noise floor of the amplifiers used making it practically impossible to record the action potential, especially on recordings that are not averaged.

4.3. Multi-channel recording technique

Two signal processing techniques were tested with the earthworm preparation and a multi-polar cuff electrode: a delay adder filter, a type of beam forming filter, and a matched filter, a type of optimal filter. The two filters were selected to enhance the detection and separatibility of action potentials based upon their conduction velocity and their shape. The delay adder filter tuned to a particular conduction velocity was found to enhance action potentials with the tuned conduction velocity and tuned direction. Unexpectedly, the unit, the LGF, generally had a larger amplitude spike. The 3 mm tripolar cuff configuration was used for the analysis, which was better matched to the wavelength of the LGF than the MGF fiber. A longer cuff would improve the performance for the faster unit, but at the expense of a much longer cuff electrode. The delay adder filter was able to nearly invert the amplitude relationship when tuned to the MGF. Although it was effective in increasing the selectivity of the recording based on unit conduction velocity, it had little effect on reducing the noise in the recording, or reducing the stimulus artifact.

The matched filter was placed in series following the delay adder filter in our experiment. It generally did not enhance the differentiation of the two units based on shape and generally attenuated the amplitude of the entire signal. However, the matched filter significantly increased the SNR of the recording. It was, moreover, able to reduce the stimulus artifact down to the level of the background noise. Although there was insufficient statistical power to indicate a significant change in the peak to peak magnitude of the background noise, there was a trend which showed a decrease in its mean peak to peak amplitude. The increase in SNR was probably a result of an enhancement of signal amplitude and reduction of background noise.

Previously, narrow band filters were used to improve velocity selectivity [13], but distortion of the signals in time will be such that side-by-side comparison may no longer be appropriate. Narrow band filters have pre-set corner frequencies whereas the passbands of the matched filters are optimized to the desired signals because the filter is constructed based on the optimal template of the signal itself. The exact passband frequencies are therefore not known beforehand but can be obtained by comparing the output of the matched filter with the input. For example, for one of the experiment passbands of approximately 300–850 Hz and 600–1600 Hz were found for respectively LGF with CV of 7.5 m/s and MGF with a CV of 17.5 m/s.

The tuning curve (Fig. 7) shows the two filters have different sensitivities. The MGF filter has sensitivity with a Full Width Half Height of approximately 10–15 m/s, while the LGF has a FWHH of approximately 5 m/s. In the case of the worm, the conduction velocities of the two units are a factor of 2 apart from one another and the sensitivities of the filters do not overlap. However, in the case of mammalian nerves, a population of fibers will have the same conduction velocity, and there will be a continuous distribution of conduction velocities, and may be problematic when looking at mass activity recordings.

5. Conclusions

The earthworm is a simple, low cost, readily available, yet surprisingly useful model of the peripheral nerve and can be used to test new electrodes, algorithms, recording and stimulation instrumentation and techniques for ultimate use in neuroprosthetics. A minimal amount of translation of the earthworm results needs to be made to convert the results to the mammalian nerve, which generally has nerve fibers with higher conduction velocities. The model is a good first step between conception of a technique or computer modeling and moving to a mammalian model.

We have used the model to test a velocity selective recording technique with a multi-contact circumferential cuff electrode. The technique cascades a delay adder filter (a simple Delay and Sum-beamformer) with a matched filter. We found the delay adder to be an effective velocity sensitive filter, able to discriminate units based on conduction velocity, and the matched filter to be an effective means to eliminate artifact and increase SNR.

Conflict of interest

The authors declare they have no conflict of interests.

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